Visual-Intertial Simultaneous Localization and Mapping

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***Abstract*—Visual-Inertial Simultaneous Localization and Mapping (VI-SLAM) is a fundamental capability for autonomous robotic navigation in GPS-denied environments. This report presents the implementation of a VI-SLAM algorithm using an Extended Kalman Filter (EKF) to jointly estimate a robot's trajectory and construct a feature map of its surroundings. The system fuses high-frequency inertial measurements with visual feature observations from a calibrated stereo camera. The IMU data, consisting of linear and angular velocities, drives the EKF prediction step based on $SE(3)$ rigid-body kinematics to track the robot's pose. Concurrently, visual features tracked across the stereo image frames provide the measurement updates using a stereo-camera observation model. By coupling these sensor modalities, the EKF effectively corrects the inherent drift in IMU dead-reckoning while building a sparse, consistent map of the environment.**

***Index Terms*—SLAM, Extended Kalman Filter, $SE(3)$, Sensor Fusion, Visual-Inertial Odometry.**

I. Introduction

Simultaneous Localization and Mapping (SLAM) is the computational problem of constructing or updating a map of an unknown environment while simultaneously keeping track of a robot's location within it. Visual-Inertial SLAM (VI-SLAM) leverages the complementary characteristics of cameras and Inertial Measurement Units (IMUs) to achieve this. While IMUs provide high-rate ego-motion estimates, they are inherently prone to significant drift over time due to the integration of sensor noise. Conversely, cameras capture rich information about the environment, allowing the system to observe static landmarks and correct accumulated drift.

Solving the VI-SLAM problem is critical for robust autonomous navigation in environments where external positioning systems are unavailable, such as indoor spaces or urban canyons. In this project, we tackle the VI-SLAM problem using an Extended Kalman Filter (EKF) framework. The state of the system encompasses both the robot's trajectory in the Special Euclidean group $SE(3)$ and a map of static 3D point landmarks $m \in \mathbb{R}^{3M}$. We utilize IMU for dead reckoning and initial trajectory estimates. Visual features extracted from the stereo camera feed serve as observations. Using the known intrinsic and extrinsic calibration of the stereo setup, we project the estimated 3D landmarks onto the optical frames. The discrepancy between the predicted and actual pixel coordinates forms the innovation for the EKF update step. This step simultaneously refines the static landmark coordinates and corrects the IMU trajectory drift, resulting in a more robust state estimate.

II. Problem Formulation

A. State Space Formulation

We will leverage IMU for the dead reckoning capabilities. For the same, we define the state of the IMU at any given time step t using a rigid-body transformation matrix, T_t, belonging to the Special Euclidean group $SE(3)$:

$$T_t = \begin{bmatrix} R_t & p_t \\ \mathbf{0}^\top & 1 \end{bmatrix} \in SE(3)$$

Here, $p_t \in \mathbb{R}^3$ denotes the world frame position of the robot, while $R_t \in SO(3)$ denotes the rotation from the world frame into the local IMU/body reference frame. To manage the uncertainty of this pose, we utilize a 6×6 covariance matrix, $\Sigma_t \in \mathbb{R}^{6\times 6}$, defined in the local tangent space (the Lie algebra $\mathfrak{se}(3)$). The six degrees of freedom within the perturbation vector are organized sequentially: the three translational components (v) are followed by the three rotational components (ω).

The environment is populated by stationary landmarks, denoted as $m_j \in \mathbb{R}^3$, which represent fixed 3D coordinates in the world frame. For the complete Visual-Inertial SLAM implementation, the system's aggregate state vector encapsulates both the current IMU trajectory and the entire map of N landmarks:

$$\xi = (T_t, m_0, \ldots, m_N)$$

B. Motion Model

At any discrete time step t, the transition to the subsequent pose x_{t+1} is governed by a motion model. This function predicts the new pose based on the previous pose x_t, the applied control inputs (or odometry measurements) u_t, and inherent system noise w_t:

$$x_{t+1} = f(x_t, u_t, w_t) \sim p_f(\cdot \mid x_t, u_t)$$

The system noise w_t is modeled deterministically for each dataset to ensure high accuracy of plots and is formulated as simple $\sigma_w I$ for scalar σ_w To realize the probabilistic

framework for rigid body navigating a aforementioned 3D state space, we use $SE(3)$ transformation.

$$\dot{T}_t = T_t \cdot \begin{bmatrix} \widehat{\omega}_t & v_t \\ \mathbf{0}^\top & 0 \end{bmatrix}$$

Here, the vectors $v_t \in \mathbb{R}^3$ and $\omega_t \in \mathbb{R}^3$ correspond to the linear and angular velocities measured directly by the IMU in its local coordinate frame. The operator $\widehat{\cdot}$ (hatmap) converts the angular velocity vector into its associated 3×3 skew-symmetric matrix.

The disparity in timestamp between any two instances is given by τ. We integrate the given kinematic model over this discrete time disparity τ, obtaining the aforementioned next step x_{t+1}.

C. Observation Model

To successfully map the environment and with-hold sensor drift, the system relies on stereo camera sensor data. At any given time step t, the detection of a static spatial landmark $\mathbf{m}_i \in \mathbb{R}^3$ generates an observation vector $\mathbf{z}_{t,i}$. We frame this as probabilistic function like the motion model, where the measurement is a nonlinear function $h(\cdot)$ of the current robot state $\mathbf{x}_t$ and the landmark's position, corrupted by zero-mean Gaussian observation noise $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \Sigma_v)$:

$$\mathbf{z}_{t,i} = h(\mathbf{x}_t, \mathbf{m}_i) + \mathbf{v}_t$$

where $\Sigma_v = \sigma_v I$, similar to the motion model noise is varied manually from dataset to dataset for improved results. The formulation of the nonlinear observation function $h(\cdot)$ requires mapping the global 3D point into the optical frame of the cameras. This is achieved by compounding the vehicle's current spatial pose with the known extrinsic calibration ${}_I T_C \in SE(3)$, which defines the fixed transformation from the body frame to the optical center.

Let $\mathbf{p}_c = [x, y, z]^\top$ denote the coordinates of the landmark $\mathbf{m}_i$ after it has been transformed into the left camera frame. To model how this 3D point is captured by the digital sensor, we apply a perspective projection utilizing the camera's intrinsic calibration matrix K.

Because the system employs a stereo camera, a single landmark observation actually yields a stacked, 4-dimensional measurement vector comprising the horizontal and vertical pixel coordinates from both the Left (L) and Right (R) cameras:

$$\mathbf{z}_{t,i} = \begin{bmatrix} u_L \\ v_L \\ u_R \\ v_R \end{bmatrix}$$

For this 4-dimensional coordinates, we construct K_s matrix representing the Intrinsic matrix of the stereo pair separated by a horizontal baseline b (which is extracted from the relative extrinsic calibrations of the two lenses), which we use along with the projection function π. The right camera observes the identical point (x, y, z) but shifted along the horizontal axis by the baseline distance. This K_s matrix is given by,

$$\begin{bmatrix} u_L \\ v_L \\ u_R \\ v_R \end{bmatrix} = \frac{1}{z} \begin{bmatrix} f_u^L & 0 & c_u^L & 0 \\ 0 & f_v^L & c_v^L & 0 \\ f_u^R & 0 & c_u^R & -f_u^R b \\ 0 & f_v^R & c_v^R & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

D. The SLAM Problem

To make the problem solvable and reduce computational overhead, we apply the Markov assumption. We assume that a robot's current pose depends only on its immediately preceding pose and the current control input, and that an observation depends solely on the current pose and the observed landmarks.

By leveraging these conditional independencies, we can factorize the complex posterior of SLAM into three distinct, probabilistic models. The first is the Prior $p(x_0)$ which represents the robot's initial state belief. Second, the Motion Model $p(x_t \mid x_{t-1}, u_t)$ which is the probability of transitioning to a new pose given the previous pose and the movement commands. Finally the Observation Model $p(z_{t,i} \mid x_t, m_i)$: The likelihood of making a specific sensor measurement given the robot's current pose and the true location of the map landmark. Combining these components yields the factorized formulation of the SLAM problem:

$$p(x_{0:T}, m \mid z_{1:T}, u_{1:T})$$

$$\propto p(x_0) \prod_{t=1}^{T} \left(p(x_t \mid x_{t-1}, u_t) \prod_i p(z_{t,i} \mid x_t, m_i) \right)$$

This decomposed equation is the foundation of SLAM.

III. Technical Approach

A. IMU Localization - EKF Prediction

The objective of the IMU localization process is to continuously estimate the rigid-body pose of the robot by integrating linear and angular velocity measurements over time. To maintain statistical consistency, we formulate this dead-reckoning process within an Extended Kalman Filter (EKF) prediction framework, operating directly on the Lie group $SE(3)$ to handle rotational and translational kinematics.

1) System State and Initialization: The discrete-time state of the system at time step k is represented by the homogeneous transformation matrix $T_k \in SE(3)$, which maps coordinates from the local IMU frame to the fixed world frame. The initial pose is anchored at the origin, $T_0 = I_{4\times4}$. The uncertainty associated with the pose estimate is tracked via a 6×6 covariance matrix Σ_k, defined in the local tangent space (the Lie algebra $\mathfrak{se}(3)$). Based on the implemented state ordering, the six degrees of freedom are arranged as translational perturbations followed by rotational perturbations: $\delta\xi = [\delta v^\top, \delta\omega^\top]^\top \in \mathbb{R}^6$. The covariance is initialized identically with a small isotropic uncertainty, $\Sigma_w = 0.001 I_6$

2) Mean Pose: At each time interval $\tau = t_{k+1} - t_k$, the IMU yields a linear velocity $v_k \in \mathbb{R}^3$ and an angular velocity $\omega_k \in \mathbb{R}^3$ in the body frame. We map these physical rates into a continuous-time spatial twist matrix $\widehat{u}_k \in \mathfrak{se}(3)$:

$$\widehat{u}_k = \begin{bmatrix} \widehat{\omega}_k & v_k \\ \mathbf{0}^\top & 0 \end{bmatrix} \in \mathbb{R}^{4\times 4}$$

where $\widehat{\omega}_k \in \mathfrak{so}(3)$ is the 3×3 skew-symmetric matrix corresponding to the angular velocity vector. To propagate the mean pose forward in time, we integrate this twist over the duration of the time step τ. We utilize matrix exponential (computed via standard linear algebra libraries) to map the Lie algebra velocity directly onto the Lie group manifold:

$$T_{k+1} = T_k \exp(\tau \cdot \widehat{u}_k)$$

3) Covariance: To accurately pass the state uncertainty Σ_k, we must transition the covariance matrix through the nonlinear $SE(3)$ dynamics. Following the theoretical EKF transition matrix requires evaluating how small perturbations in the local tangent frame evolve over the period. We construct the 6×6 adjoint operator matrix of the Lie algebra, $\mathrm{ad}_{u_k} \in \mathbb{R}^{6\times 6}$, which dictates the differential twist:

$$\mathrm{ad}_{u_k} = \begin{bmatrix} \widehat{\omega}_k & \widehat{v}_k \\ \mathbf{0}_{3\times 3} & \widehat{\omega}_k \end{bmatrix}$$

Using this adjoint, the discrete-time state transition matrix Ψ_k is derived by taking the matrix exponential of the scaled, negated adjoint operator:

$$\Psi_k = \exp(-\tau \cdot \mathrm{ad}_{u_k})$$

Applying this linear transition mapping, the predicted covariance for the next time step becomes:

$$\Sigma_{k+1} = \Psi_k \Sigma_k \Psi_k^\top + \tau W$$

This formulation exactly propagates the existing uncertainty through the nonlinear rotation dynamics while injecting the scaled motion noise (τW) accumulated over the discrete integration step.

B. Landmark Mapping - EKF Update

In this phase of the estimation pipeline, the IMU trajectory derived from the prediction step is treated as known and fixed. The objective is strictly map the environment by estimating the 3D spatial coordinates of static landmarks and their associated uncertainties, conditioned on the fixed trajectory.

1) Stereo Triangulation and Landmark Initialization: When a tracked feature i is observed for the first time at time step t, its 3D position is initialized directly via stereo triangulation. Given the left pixel coordinates (u_L, v_L) and the right horizontal pixel u_R, the horizontal stereo disparity is computed. To ensure numerical stability and prevent division by zero for points near infinity, a heuristic threshold is applied to the disparity (clipping values below 0.5 pixels).

The depth z_c in the left camera's optical frame is recovered using the focal length and the stereo baseline b:

$$z_c = \frac{f_u \cdot b}{(u_L - c_u^L) - (u_R - c_u^R)}$$

The remaining spatial coordinates x_c and y_c are derived via the standard pinhole inverse projection model. This local point $\mathbf{q}_{cam} = [x_c, y_c, z_c, 1]^\top$ is then transformed into the world frame using the current IMU pose T_t and the known IMU-to-Camera extrinsic transformation T_{imu_cam}:

$$\mathbf{m}_i = T_t \cdot T_{imu_cam} \cdot \mathbf{q}_{cam}$$

Upon initialization, the covariance block for the new landmark set with isotropic variance $(0.05 \cdot I_3)$.

2) Jacobians and Measurements: For landmarks that have been initialized, the filter performs a corrective update based on re-observation. The active world-frame landmark means μ_{obs} are first transformed back into the current local camera frame. The 3D points are then passed through a nonlinear perspective projection function $\pi(\cdot)$ and multiplied by the stacked stereo intrinsic matrix K_s to generate the 4D predicted measurement vector $\tilde{z}$.

To linearize this measurement model for the EKF, the analytical observation Jacobian $H_i \in \mathbb{R}^{4\times 3}$ is computed for each landmark using the chain rule. It combines the derivative of the perspective projection with respect to the camera-frame coordinates $(\frac{\partial \pi}{\partial \mathbf{q}})$ and the spatial transformation matrix mapping world coordinates to the camera frame:

$$H_i = K_s \cdot \frac{\partial \pi}{\partial \mathbf{q}} \cdot (T_{imu_cam} \cdot T_t^{-1}) \cdot P_T$$

where $P_T \in \mathbb{R}^{4\times 3}$ is a projection matrix that simply extracts the x, y, z components. To process all currently visible landmarks efficiently in a single vectorized operation, these individual Jacobians are populated into a large, sparse block-diagonal matrix H_{sparse}.

3) EKF Update Formulation: With the sparse measurement Jacobian constructed, the filter executes the standard EKF update cycle for all actively observed landmarks simultaneously. The measurement noise covariance V is expanded into a block-diagonal sparse matrix corresponding to the number of active observations.The innovation covariance S and the Kalman gain K are computed using sparse matrix algebra:

$$S = H_{sparse} \Sigma_{active} H_{sparse}^\top + V_{kron}$$

$$K = \Sigma_{active} H_{sparse}^\top S^{-1}$$

Finally, the state mean and the active sub-block of the covariance matrix are updated using the innovation, where z_{meas} represents the actual stacked pixel observations from the stereo image:

$$\mu_{active} \leftarrow \mu_{active} + K(z_{meas} - \tilde{z})$$

$$\Sigma_{active} \leftarrow (I_3 - K H_{sparse}) \Sigma_{active}$$

C. Visual Inertial Simultaneous Localization & Mapping

In this final phase of the estimation pipeline, the previous decoupled approaches are synthesized into a coupled VI-SLAM framework. Rather than treating the trajectory as known or ignoring map correlations, the robot's pose and the environment map are estimated simultaneously. This requires maintaining the statistical interdependencies between the IMU trajectory and every mapped landmark. This means the state is expanded into a single, unified representation. The mean state consists of the current IMU pose $T_t \in SE(3)$ and a flattened vector of all M 3D landmark coordinates, $\mu_{map} \in \mathbb{R}^{3M}$. To capture the coupled uncertainty, the system maintains a joint covariance matrix $\Sigma \in \mathbb{R}^{(6+3M)\times(6+3M)}$.

The top-left 6×6 block represents the IMU pose uncertainty in the local tangent space, initialized to $0.001\cdot I_6$. The diagonal 3×3 blocks extending downwards represent individual landmark uncertainties, while the off-diagonal blocks capture the critical cross-covariances (pose-to-landmark and landmark-to-landmark correlations).

1) EKF Prediction with Cross-Covariance: During the prediction step, the continuous-time linear and angular velocities from the IMU are integrated over the discrete time interval τ. The mean pose is propagated forward via the closed-form matrix exponential in the local body frame:

$$T_{t+1} = T_t \exp(\tau \cdot \widehat{u}_t)$$

Unlike simplified formulations that freeze cross-covariance terms, the implemented architecture strictly uses both the pose uncertainty and the pose-landmark cross-covariances through the nonlinear system dynamics. Let $F_{pose} = \exp(-\tau \cdot \mathrm{ad}_{u_t}) \in \mathbb{R}^{6\times6}$ be the discretized state transition matrix derived from the Lie algebra adjoint. The prediction step updates the joint covariance as follows:

$$\Sigma_{[:6,:6]} \leftarrow F_{pose}\Sigma_{[:6,:6]}F_{pose}^\top + \tau W$$

$$\Sigma_{[:6,6:]} \leftarrow F_{pose}\Sigma_{[:6,6:]}$$

$$\Sigma_{[6:,:6]} \leftarrow \Sigma_{[:6,6:]}^\top$$

where W is the injected 6×6 process noise covariance. The landmark-only covariance blocks remain unaffected during the IMU-driven prediction.

2) Joint Measurement Jacobians: For a batch of N active observations, the system constructs a dense joint Jacobian $H_{joint} \in \mathbb{R}^{4N\times(6+3N)}$. This matrix is formed by horizontally stacking the pose Jacobian H_{pose} and the landmark Jacobian $H_{landmark}$. Pose Jacobian (H_{pose}) is used to compute the derivative of the measurement with respect to the IMU pose, the implementation utilizes a local (right) perturbation formulation on $SE(3)$. Let $\mathbf{s}_{imu}$ be the 3D landmark coordinates transformed into the IMU frame. The derivative of this spatial transformation with respect to a local twist is handled via the $\odot$ (odot) operator, which maps a 3D point to a 4×6 generator matrix. The resulting pose Jacobian for a single observation evaluates to:

$$H_{pose_i} = -K_s \cdot \frac{\partial\pi}{\partial\mathbf{q}} \cdot T_{cam_imu} \cdot \mathbf{s}_{imu}^{\odot}$$

3) EKF Update: The system extracts the active subset of the covariance matrix, Σ_{active}, which contains the 6×6 pose block, the relevant 3×3 landmark blocks, and their corresponding cross-terms. The innovation covariance S and the full-state Kalman gain K_{full} are computed using the joint Jacobian:

$$S = H_{joint}\Sigma_{active}H_{joint}^\top + V$$

$$K_{full} = \Sigma_{[:,active]}H_{joint}^\top S^{-1}$$

where V is the expanded block-diagonal measurement noise matrix. The resulting state update vector $\delta \in \mathbb{R}^{6+3N}$ dictates the corrections. Because the pose Jacobian was derived using a local frame perturbation, the 6-DOF pose correction $\delta_{[:6]}$ is applied to the transformation matrix via right-multiplication, preserving the Lie group manifold structure:

$$T_t \leftarrow T_t \exp(\widehat{\delta}_{[:6]})$$

Simultaneously, the active landmark means are updated via standard Euclidean addition ($\mu_{map} \leftarrow \mu_{map} + \delta_{[6:]}$), and the entire joint covariance matrix is updated to reflect the updated bounds and newly correlated constraints:

$$\Sigma \leftarrow \Sigma - K_{full}H_{joint}\Sigma_{[active,:]}$$

IV. Results

The estimation pipeline was evaluated across three distinct outdoor sequences (dataset_00, dataset_01, and dataset_02). The following section details the performance of the pure dead-reckoning system before any visual corrections are applied.

A. IMU Trajectories

1) IMU-Only Trajectories: The standalone IMU localization relies entirely on integrating linear and angular velocities over time. The resulting 2D trajectories for each dataset vividly illustrate both the stability of the integration method and the inevitable accumulation of sensor drift.

Dataset 00: As observed in the first plot, this sequence maps a long, sweeping arc that progresses steadily along the positive x-axis. Spanning 3,755 time steps, the path extends roughly 150 meters outward.

Dataset 01: The second plot depicts a highly dynamic, meandering counter-clockwise loop. Over 3,601 steps, the sensor travels through an area spanning approximately 100×60 meters. Because pure IMU integration continuously accumulates rotational error, the trajectory completely fails to close the loop based on the video footage, leaving an accumulated translational drift

Dataset 02: The final plot illustrates the longest sequence (6,393 steps), which takes the form of a large, near-rectangular block circuit. The path travels right, sharply up, and then proceeds across a long straightaway to the left.

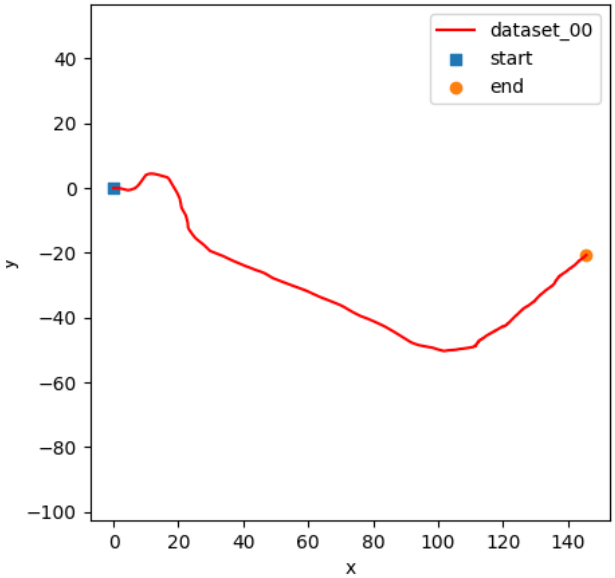

Fig. 1. IMU Trajectory: Dataset 00

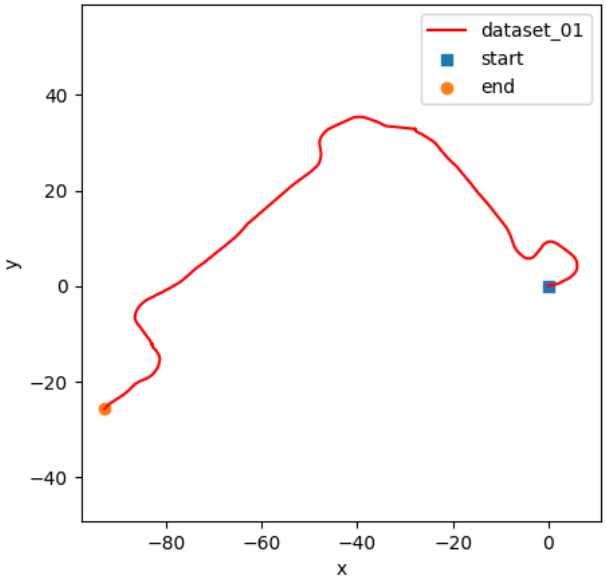

Fig. 2. IMU Trajectory: Dataset 01

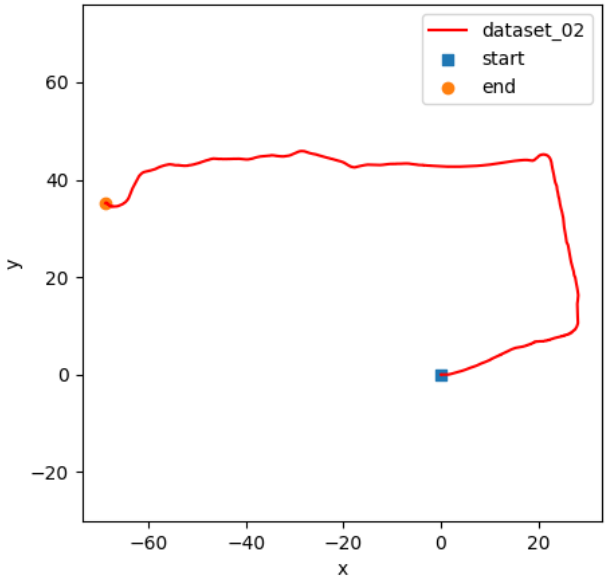

Fig. 3. IMU Trajectory: Dataset 02

2) Landmark Mapping: The second phase of the evaluation assesses the system's ability to reconstruct the environment using the fixed IMU trajectories derived in Task 1. The generated plots for all datasets overlay the estimated 2D landmark positions onto the dead-reckoned paths. Across all three sequences, the system successfully triangulates and tracks thousands of features. The resulting point clouds densely and logically populate the immediate vicinity of the robot's trajectory, matching the expected field of view of the stereo rig. This confirms the intrinsic and extrinsic transformations, along with the EKF measurement updates, are functioning correctly to map the local space.

A prominent feature in all three plots is the scattering of sparse, highly distant outliers. Also while the landmarks are locally consistent with the camera observations, their absolute global coordinates inherit the compounding translational error of the IMU. This inherited warping and the presence of observation noise directly set the stage for Task 4, where jointly optimizing the pose and the map will allow the visual observations to pull the trajectory back into alignment.

3) VI-SLAM: The final phase of the pipeline implements the fully coupled Visual-Inertial SLAM, but as evidenced by the highly jagged and erratic trajectories across all three dataset plots, the current implementation struggles. A primary factor contributing to this chaotic state estimation is the lack of robust feature selection and outlier rejection.

Furthermore, the plots vividly illustrate the severe consequences of incorrect covariance weightings. The sensitive nature of the updated trajectories suggests a fundamental imbalance in the filter's trust between the IMU dead-reckoning and the visual observations. Specifically, the measurement noise parameters are likely set too aggressively, causing the filter to over-weight noisy pixel data while under-weighting

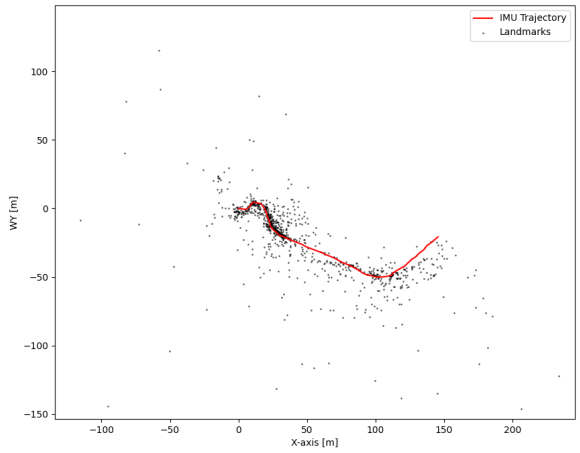

Fig. 4. Landmark Mapping: Dataset 00

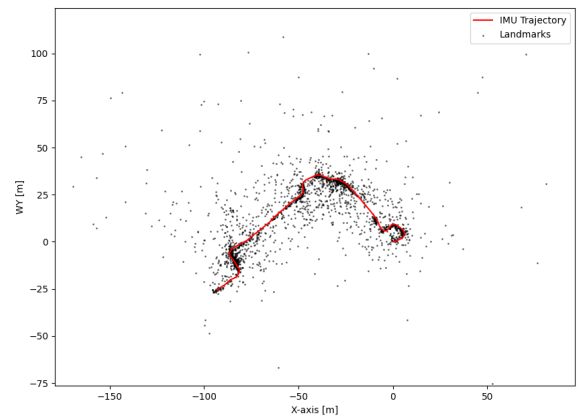

Fig. 5. Landmark Mapping: Dataset 01

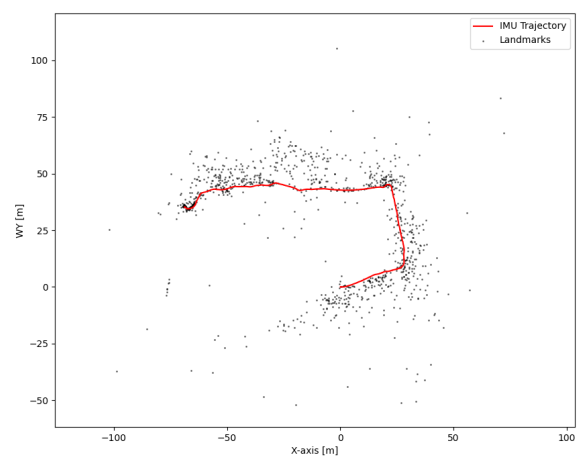

Fig. 6. Landmark Mapping: Dataset 02

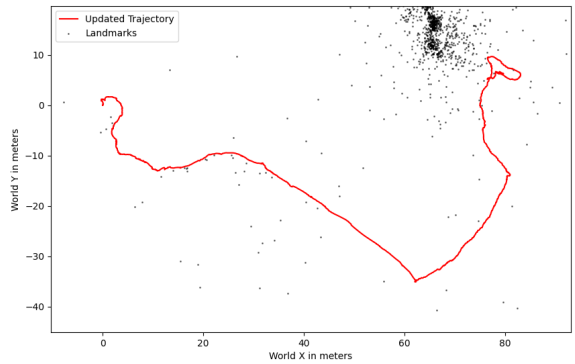

Fig. 7. Full Visual Inertial SLAM: Dataset: 00

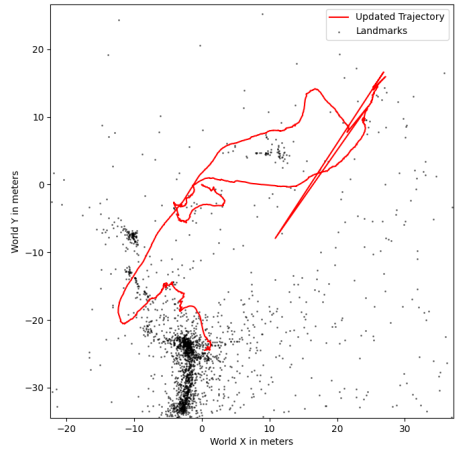

Fig. 8. Full Visual Inertial SLAM: Dataset: 01

the inherently smooth kinematic priors provided by the IMU process noise. Achieving a stable, accurate VI-SLAM system strictly requires rigorous weight tuning to properly balance these covariances, ensuring the filter correctly anchors the drift without violently snapping the trajectory.

V. Conclusion

This report detailed the development and evaluation of an Extended Kalman Filter-based Visual-Inertial SLAM system. The foundational mathematical models for rigid-body kinematics and stereo perspective projection were successfully validated. The closed-form matrix exponential stably integrated high-frequency IMU data over thousands of steps, while the stereo triangulation model effectively initialized 3D environmental features.

However, the experimental results underscored the inherent limitations of isolated sensor modalities and the sensitivity to noise tuning. While the IMU-only trajectories demonstrated

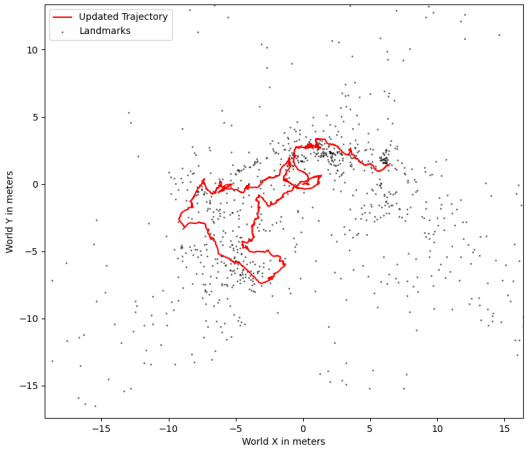

Fig. 9. Full Visual Inertial SLAM: Dataset: 02

smooth local kinematics, they inevitably accumulated significant translational drift over time. Conversely, the fully coupled VI-SLAM implementation exposed the critical necessity for increased fine-tuning and feature modeling. Ultimately, achieving a reliable and stable visual-inertial state estimator requires not just mathematically sound update equations, but rigorous fine-tuning, cross validation and extensive research into outlier rejection.